

Étale Covering Space

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Theorem.(Fundamental Theorem of Galois Theory)

If F is a finite dimensional Galois extension of K , then there is a bijection:

$\{\text{Intermediate fields of } F/K\}$



$\{\text{Subgroups of } \text{Gal}(F/K)\}.$

Besides,

$$|\text{Gal}(F/K)| = [F : K].$$

Connection between Galois Theory and Algebraic topology

Review of algebraic topology.

Definition.(Covering space)

We firstly define the covering map. Let $p : X \rightarrow S$ satisfy the following condition:

Each point $s \in S$ has an open neighborhood U in S such that $p^{-1}(U)$ is a union of disjoint open sets in X , each of which is mapped homeomorphically onto U by p .

A covering space of a space S is a space X together with a covering map.

Connection between Galois Theory and Algebraic topology

Review of algebraic topology.

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A covering space of a space S is a space X together with a covering map.

Definition.

$\forall s \in S$, say $p^{-1}(s)$ the fiber of s , and say $|p^{-1}(s)|$ the sheet of the covering which is not depended by the choice of s . Then we have a group action $\pi_1(S, s) \curvearrowright p^{-1}(s)$.

Proposition.

The induced map of the fundamental groups

$$p_\pi : \pi_1(X, x) \rightarrow \pi_1(S, s)$$

is injective. Define $H_e := p_\pi(\pi_1(X, x))$.

Connection between Galois Theory and Algebraic topology

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Proposition.

$[\pi_1(S, s) : H_e]$ = the sheet of p .

Recall the fact in Galois theory:

$$|Gal(F/K)| = [F : K].$$

Theorem.

Let S be path-connected, locally path-connected, and semi-locally simply-connected. Then there is a bijection :

$$\{\text{Path-connected covering space}\}$$

$$\{\text{Subgroup of } \pi_1(S, s)\}.$$

Recall the Fundamental Theorem of Galois theory.

Connection between Galois Theory and Algebraic topology

Definition.(Universal covering)

A simply-connected covering space of S is called a universal cover, say $\tilde{S}$. It is unique up to isomorphism.

In Galois theory, we have "Algebraic Closure".

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Proposition.

$\text{Aut}(\tilde{S}/S) \cong \pi_1(S, s)$. Hence $\text{Aut}(\tilde{S}/S)$ can be viewed as a second definition of the fundamental group of S .

In Galois theory, we have "absolute Galois group".

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Definition.(Sheaf)

Let X be a topological space. A sheaf $\mathcal{F}$ of abelian groups on X consists of the data:

- (a) for every open subset $U \subseteq X$, an abelian group $\mathcal{F}(U)$,
- (b) for every inclusion $V \subseteq U$ of open subsets of X , a morphism of abelian groups $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$,

subject to the conditions

- (0) $\mathcal{F}(\emptyset) = 0$, where $\emptyset$ is the empty set,
- (1) ρ_{UU} is the identity map $\mathcal{F}(U) \rightarrow \mathcal{F}(U)$, and
- (2) if $W \subseteq V \subseteq U$ are three open subsets, then $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$.
- (3) if U is an open set, if $\{V_i\}$ is an open covering of U , and if $s \in \mathcal{F}(U)$ is an element such that $s|_{V_i} = 0$ for all i , then $s = 0$;
- (4) if U is an open set, if $\{V_i\}$ is an open covering of U , and if we have elements $s_i \in \mathcal{F}(V_i)$ for each i , with the property that for each i, j , $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$, then there is an element $s \in \mathcal{F}(U)$ such that $s|_{V_i} = s_i$ for each i .

Definition.(Stalk)

If $\mathcal{F}$ is a sheaf on X , and if P is a point of X , we define the stalk $\mathcal{F}_P$ of $\mathcal{F}$ at P to be the direct limit of the groups $\mathcal{F}(U)$ for all open sets U containing P , via the restriction maps ρ . Namely

$$\varinjlim_{P \in U} \mathcal{F}(U).$$

Definition. (Morphisms between sheaves on X)

If $\mathcal{F}$ and $\mathcal{G}$ are sheaves on X , a morphism $\varphi : \mathcal{F} \rightarrow \mathcal{G}$ consists of a morphism of abelian groups $\varphi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for each open set U , such that whenever $V \subseteq U$ is an inclusion, and the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi(U)} & \mathcal{G}(U) \\ \downarrow \rho_{UV} & & \downarrow \rho'_{UV} \\ \mathcal{F}(V) & \xrightarrow{\varphi(V)} & \mathcal{G}(V) \end{array}$$

is commutative, where ρ and ρ' are the restriction maps in $\mathcal{F}$ and $\mathcal{G}$. If $\mathcal{F}$ and $\mathcal{G}$ are sheaves on X .

Definition.(Spectrum)

- (1) As a set, we define $\text{Spec } A$ to be the set of all prime ideals of A .
 - (2) As a topological space. If $\mathfrak{a}$ is any ideal of A , we define the subset $V(\mathfrak{a}) \subseteq \text{Spec } A$ to be the set of all prime ideals which contain $\mathfrak{a}$. Let the subsets of the form $V(\mathfrak{a})$ to be the closed subsets. Note that $V(A) = \emptyset$; $V((0)) = \text{Spec } A$.
 - (3) A sheaf of rings $\mathcal{O}$ on $\text{Spec } A$. For each prime ideal $\mathfrak{p} \subseteq A$, let $A_{\mathfrak{p}}$ be the localization of A at $\mathfrak{p}$. For an open set $U \subseteq \text{Spec } A$, we define $\mathcal{O}(U)$ to be the set of functions $s : U \rightarrow \prod_{\mathfrak{p} \in U} A_{\mathfrak{p}}$, such that $s(\mathfrak{p}) \in A_{\mathfrak{p}}$ for each $\mathfrak{p}$, and such that s is locally a quotient of elements of A : to be precise, we require that for each $\mathfrak{p} \in U$, there is a neighborhood V of $\mathfrak{p}$, contained in U , and elements $a, f \in A$, such that for each $\mathfrak{q} \in V$, $f \notin \mathfrak{q}$, and $s(\mathfrak{q}) = a/f$ in $A_{\mathfrak{q}}$.
- Let A be a ring. The spectrum of A is the pair consisting of the topological space $\text{Spec } A$ together with the sheaf of rings $\mathcal{O}$ defined above.

Definition. (Ringed space)

A ringed space is a pair $(X, \mathcal{O}_X)$ consisting of a topological space X and a sheaf of rings $\mathcal{O}_X$ on X . A morphism of ringed spaces from $(X, \mathcal{O}_X)$ to $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$ of a continuous map $f : X \rightarrow Y$ and a map $f^\# : \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$ of sheaves of rings on Y . The ringed space $(X, \mathcal{O}_X)$ is a locally ringed space if for each point $P \in X$, the stalk $\mathcal{O}_{X,P}$ is a local ring.

Definition. (Affine scheme)

An affine scheme is a locally ringed space which is isomorphic to spectrum of some ring.

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Definition.(Scheme)

A scheme is a locally ringed space $(X, \mathcal{O}_X)$ in which every point has an open neighborhood U such that the topological space U , together with the restricted sheaf $\mathcal{O}_X|_U$, is an affine scheme.

We call X the underlying topological space of the scheme $(X, \mathcal{O}_X)$, and $\mathcal{O}_X$ its structure sheaf.

A morphism of schemes is a morphism as locally ringed space.

Algebraic Geometry

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A morphism of schemes is a morphism as locally ringed space.

Example.

If k is a field, $\text{Spec } k$ is an affine scheme whose topological space consists of one point, and whose structure sheaf consists of the field k .

Definition.(Connected scheme)

A scheme is connected if its underlying topological space is connected.

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Definition.(Finite morphism)

A morphism $f : X \rightarrow Y$ is a finite morphism if there exists a covering of Y by open affine subsets $V_i = \text{Spec } B_i$, such that for each i , $f^{-1}(V_i)$ is affine, equal to $\text{Spec } A_i$, where A_i is a B_i -algebra which is a finitely generated B_i -module.

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Theorem.

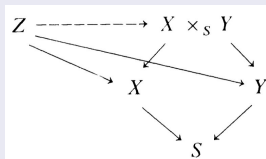
Finite morphisms are closed maps.

Definition.(Open immersion)

An open subscheme of a scheme X is a scheme U , whose topological space is an open subset of X , and whose structure sheaf $\mathcal{O}_U$ is isomorphic to the restriction $\mathcal{O}_X|_U$ of the structure sheaf of X . An open immersion is a morphism $f : X \rightarrow Y$ which induces an isomorphism of X with an open subscheme of Y .

Definition.(Fibred product)

Let S be a scheme, and let X, Y be schemes over S , i.e., schemes with morphisms to S . We define the fibred product of X and Y over S , denoted $X \times_S Y$, to be a scheme, together with morphisms $p_1 : X \times_S Y \rightarrow X$ and $p_2 : X \times_S Y \rightarrow Y$, which make a commutative diagram with the given morphisms $X \rightarrow S$ and $Y \rightarrow S$, such that given any scheme Z over S , and given morphisms $f : Z \rightarrow X$ and $g : Z \rightarrow Y$ which make a commutative diagram with the given morphisms $X \rightarrow S$ and $Y \rightarrow S$, then there exists a unique morphism $\theta : Z \rightarrow X \times_S Y$ such that $f = p_1 \circ \theta$, and $g = p_2 \circ \theta$. The morphisms p_1 and p_2 are called the projection morphisms of the fibred product onto its factors.



Theorem.

For any two schemes X and Y over a scheme S , the fibred product $X \times_S Y$ exists, and is unique up to unique isomorphism.

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Definition.(Base change)

Let S be a fixed scheme which we think of as a base scheme, meaning that we are interested in the category of schemes over S . For example, think of $S = \operatorname{Spec} k$, where k is a field. If S' is another base scheme, and if $S' \rightarrow S$ is a morphism, then for any scheme X over S , we let $X' = X \times_S S'$, which will be a scheme over S' . We say that X' is obtained from X by making a base change $S' \rightarrow S$.

If a property of morphisms is also held for its base change, we say that this property is stable under base change.

Definition. (Sheaf of modules)

Let $(X, \mathcal{O}_X)$ be a ringed space. A sheaf of $\mathcal{O}_X$ -modules (or simply an $\mathcal{O}_X$ -module) is a sheaf $\mathcal{F}$ on X , such that for each open set $U \subseteq X$, the group $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module, and for each inclusion of open sets $V \subseteq U$, the restriction homomorphism $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is compatible with the module structures via the ring homomorphism $\mathcal{O}_X(U) \rightarrow \mathcal{O}_X(V)$. A morphism $\mathcal{F} \rightarrow \mathcal{G}$ of sheaves of $\mathcal{O}_X$ -modules is a morphism of sheaves, such that for each open set $U \subseteq X$, the map $\mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is a homomorphism of $\mathcal{O}_X(U)$ -modules.

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Definition. (Flat module)

Let A be a ring. An A -module M is called flat if the functor $N \mapsto M \otimes_A N$ on the category of A -modules is exact.

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Definition.(Flat morphism)

A morphism $f : X \rightarrow Y$ of schemes is called flat if $\forall x \in X$, $\mathcal{O}_{X,x}$ is flat over $\mathcal{O}_{Y,f(x)}$.

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Definition.(Unramified morphism)

Let $f : X \rightarrow Y$ be a morphism locally of finite type, x a point in X , and $y = f(x)$. We say f is unramified if $\mathfrak{m}_y \mathcal{O}_{X,x} = \mathfrak{m}_x$ and the residue field $k(x)$ is a finite separable extension of the residue field $k(y)$. (Consider algebraic number theory)

Definition. (Étale morphism)

If f is a morphism locally of finite presentation, we say f is étale if f is flat and unramified.

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Theorem.

The following properties of morphisms are stable under base change.

- (1) Finite.
- (2) Affine.
- (3) Flat.
- (4) Étale.
- (5) Locally free of finite rank.

Proposition.

In the following, morphisms are assumed to be locally of finite presentation.

- (1) Composites of étale morphisms are étale.
- (2) Open immersions are étale.
- (3) Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two morphisms such that gf is étale and g is étale. Then f is étale.
- (4) Étale morphisms are open mappings.

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- (4) Étale morphisms are open mappings.

Proof.

Let us prove (3) as an example. By 2.2.4 in [1], the graph $\Gamma_f : X \rightarrow X \times_Z Y$ of f is an open immersion, which is étale by (2). Since étale is stable under base change, the projection $p_2 : X \times_Z Y \rightarrow Y$ is étale. By (1), $f = p_2 \Gamma_f$ is étale.

[1]Fu Lei. Etale Cohomology Theory[M].WORLD SCIENTIFIC:2011-01.

Definition. (X/G)

Let X be a scheme on which a finite group G acts on the right. If $X = \operatorname{Spec} A$, this is equivalent to saying G acts on A on the left. Let $\operatorname{Hom}(X, Z)^G$ be the subset of $\operatorname{Hom}(X, Z)$ consisting of morphisms invariant under G . A morphism is called the quotient of X by G if it is a morphism $X \rightarrow Y$ is invariant under the action of G , such that the canonical map

$$\operatorname{Hom}(Y, Z) \rightarrow \operatorname{Hom}(X, Z)^G$$

is bijective for any scheme Z , that is, Y represents the functor $Z \mapsto \operatorname{Hom}(X, Z)^G$. We often denote Y by X/G .

Definition.(Étale Covering)

An étale covering space of a scheme S is a finite étale morphism $X \rightarrow S$.

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An étale covering space of a scheme S is a finite étale morphism $X \rightarrow S$.

The group $\text{Aut}(X/S)$ acts on X on the left. Let $G = \text{Aut}(X/S)^\circ$ be the opposite group, then G acts on X on the right. If $S = X/G$, we say $X \rightarrow S$ is a Galois étale covering space with Galois group G .

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Remark.

If X is a variety over an algebraically closed field and $\pi : Y \rightarrow X$ is finite and étale, then each fibre of π has exactly the same number of points. Moreover, each $x \in X$ has an étale neighborhood $(U, u) \rightarrow (X, x)$ such that $Y \times_X U$ is a disjoint union of open subvarieties (or subschemes) U_i ; each of which is mapped isomorphically onto U by $\pi \times 1$. Thus, a finite étale map is the natural analogue of a finite covering space in the theory of algebraic topology.

Definition. (Geometric point)

Let X be a scheme. A geometric point of X is a morphism $\gamma : s \rightarrow X$ such that s is the spectrum of a separably closed field. We often write $k(s)$ for the separably closed field such that $s = \operatorname{Spec} k(s)$. (Giving a geometric point is equivalent to giving a separably closed extension $k(s)$ of the residue field $k(x)$). Let $f : X \rightarrow X'$ be a morphism. Then $f\gamma : s \rightarrow X'$ is a geometric point of X' .

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Definition. (Pointed scheme)

A pointed scheme is a pair (X, γ) such that X is a scheme, and $\gamma : s \rightarrow X$ is a geometric point of X . A morphism $f : (X, \gamma) \rightarrow (X', \gamma')$ between pointed schemes is a morphism $f : X \rightarrow X'$ such that $f\gamma = \gamma'$.

Theorem.1

Let (S, γ) be a pointed scheme, and let $(X_i, \alpha_i) \rightarrow (S, \gamma) (i = 1, 2)$ be two morphisms of pointed schemes.

(1) If X_1 is connected and X_2 is unramified and separated over S , then there exists at most one S -morphism from (X_1, α_1) to (X_2, α_2) .

(2) Suppose X_1 and X_2 are étale covering spaces of S . There exists a connected pointed étale covering space (X_3, α_3) of (S, γ) dominating $(X_i, \alpha_i) (i = 1, 2)$, that is, there exist S -morphisms from (X_3, α_3) to (X_i, α_i) .

Proof.

- (1) An S -morphism f from X_1 to X_2 is completely determined by its graph $\Gamma_f : X_1 \rightarrow X_1 \times_S X_2$, which is a section of the projection $\pi_1 : X_1 \times_S X_2 \rightarrow X_1$. If $f(\alpha_1) = \alpha_2$, then $\Gamma_f(\alpha_1) = (\alpha_1, \alpha_2)$. Such Γ_f is unique if it exists by [1]2.3.10(i) which is a property of étale morphism.
- (2) α_1 and α_2 define a point $\alpha_3 = (\alpha_1, \alpha_2)$ of $X_1 \times_S X_2$. Let X_3 be the connected component of $X_1 \times_S X_2$ containing the image of α_3 . Then (X_3, α_3) dominates (X_i, α_i) ($i = 1, 2$).

Theorem.2

Let (S, γ) be a pointed connected noetherian scheme, X_1 and X_2 two étale covering spaces of S , $u : X_1 \rightarrow X_2$ an S -morphism, and $X_i(\gamma) (i = 1, 2)$ the sets of geometric points of X_i lying above γ . If the map $X_1(\gamma) \rightarrow X_2(\gamma)$ induced by u is bijective, then u is an isomorphism.

Theorem.2

Let (S, γ) be a pointed connected noetherian scheme, X_1 and X_2 two étale covering spaces of S , $u : X_1 \rightarrow X_2$ an S -morphism, and $X_i(\gamma) (i = 1, 2)$ the sets of geometric points of X_i lying above γ . If the map $X_1(\gamma) \rightarrow X_2(\gamma)$ induced by u is bijective, then u is an isomorphism.

Sketch of proof.

The image of any connected component of X_2 in S is both open and closed. Since S is connected, the image is S . Replacing X_2 by its connected components and X_1 by the inverse images of these components, we are reduced to the case where X_2 is connected. Note that u is finite and étale. So $u_* \mathcal{O}_{X_1}$ is a locally free $\mathcal{O}_{X_2}$ -module of constant finite rank.

Let $s \in S$ be the image of γ and let $x_2 \in X_2$ be a point above s . Then

$$X_1 \times_{X_2} \operatorname{Spec} \mathcal{O}_{X_2, x_2} \cong \operatorname{Spec} A$$

for some $\mathcal{O}_{X_2, x_2}$ -algebra A which is free of finite rank as an $\mathcal{O}_{X_2, x_2}$ -module. Since $X_1(\gamma) \rightarrow X_2(\gamma)$ is bijective, there is one and only one point x_1 in X_1 lying above x_2 . So A is a local ring. Since u is étale, $\mathfrak{m}_{x_2} A$ is the maximal ideal of A and $A/\mathfrak{m}_{x_2} A$ is finite separable over $\mathcal{O}_{X_2, x_2}/\mathfrak{m}_{x_2}$. Again because $X_1(\gamma) \rightarrow X_2(\gamma)$ is bijective, we must have $\mathcal{O}_{X_2, x_2}/\mathfrak{m}_{x_2} \cong A/\mathfrak{m}_2 A$. It follows that the rank of $u_* \mathcal{O}_{X_1}$ is 1. Let x'_2 be an arbitrary point of X_2 and let A' be an $\mathcal{O}_{X_2, x'_2}$ -algebra such that

$$X_1 \times_{X_2} \operatorname{Spec} \mathcal{O}_{X_2, x'_2} \cong \operatorname{Spec} A'.$$

Since $\operatorname{rank}(u_* \mathcal{O}_{X_1}) = 1$, A' is a free $\mathcal{O}_{X_2, x'_2}$ -module of rank 1.

The homomorphism $\mathcal{O}_{X_2, x'_2} / \mathfrak{m}_{x'_2} \rightarrow A' / \mathfrak{m}_{x'_2} A$ is a nonzero homomorphism of one dimensional vectors spaces. It is necessarily surjective. By Nakayama's lemma, the homomorphism $\mathcal{O}_{X_2, x'_2} \rightarrow A'$ is also surjective. It is injective since it is faithfully flat. So we have $\mathcal{O}_{X_2, x'_2} \cong A'$. Hence $\mathcal{O}_{X_2} \cong u_* \mathcal{O}_{X_1}$, and u is an isomorphism. The proof is complete.

Theorem.3

Let (S, γ) be a pointed connected noetherian scheme, X a connected étale covering space of S , $X(\gamma)$ the set of geometric points in X lying above γ , and $G = \text{Aut}(X/S)^\circ$. The following conditions are equivalent:

- (1) $X/G \cong S$, that is, X is a Galois covering of S .
- (2) G acts transitively on $X(\gamma)$.

For the proof, see [1]3.2.8.

Theorem.4

let S be a connected noetherian scheme and let X be a connected étale covering space of S . Then any S -morphism $u : X \rightarrow X$ is an isomorphism.

Theorem.4

let S be a connected noetherian scheme and let X be a connected étale covering space of S . Then any S -morphism $u : X \rightarrow X$ is an isomorphism.

Proof.

Note that u is necessarily étale and finite. So $u(X)$ is both open and closed. As X is connected, we have $u(X)=X$. Let γ be a geometric point of S , let α be a geometric point of X lying above γ , and let x be the image of α . Then there exists $x' \in X$ such that $u(x') = x$. Moreover the residue field $k(x')$ is a finite separable extension of the residue field $k(x)$. So there exists a geometric point α' of X with image x' such that $u\alpha' = \alpha$. Thus u induces a surjective map $X(\gamma) \rightarrow X(\gamma)$. As $X(\gamma)$ is finite. This map is bijective. By Theorem 2, u is an isomorphism.

Theorem.5

Let (S, γ) be a pointed connected noetherian scheme, and let (Y, β) be a pointed étale covering space of (S, γ) . There exists a pointed connected Galois étale covering space (X, α) of (S, γ) dominating (Y, β) , that is, there exists an S -morphism from (X, α) to (Y, β) .

Proof.

Let $\beta_1, \dots, \beta_k$ be all the distinct geometric points of Y lying above γ . They define a geometric point α of $Y^k = Y \times_S \cdots \times_S Y$ such that $p_i \alpha = \beta_i (i = 1, \dots, k)$, where $p_i : Y^k \rightarrow Y$ are the projections. Let X be the connected component of Y^k containing the image of α . It suffices to show that X is a Galois covering of S . By Theorem 3 and Theorem 4, it suffices to show that for any geometric point α' of X lying above γ , there exists an S -morphism $u : X \rightarrow X$ such that $u\alpha = \alpha'$.

Let $j : X \hookrightarrow Y^k$ be the open immersion. Since $p_i j \alpha = \beta_i (i = 1, \dots, k)$ are distinct, $p_i j \alpha'$ are distinct by Theorem 1(1). Let σ be a permutation of $\{1, \dots, k\}$ such that $p_i j \alpha' = p_{\sigma(i)} j \alpha$, and let $v : Y^k \rightarrow Y^k$ be the S-morphism with the property $p_i v = p_{\sigma(i)}$. We have

$$p_i v j \alpha = p_{\sigma(i)} j \alpha = p_i j \alpha'.$$

It follows that $v j \alpha = j \alpha'$. Since X is the connected component of Y^k containing the images of α and α' , v induces a morphism $u : X \rightarrow X$ with the property $u(\alpha) = \alpha'$. This proves our assertion.

Recall some homological algebra.

Proposition.

Consider the following conditions of the category I :

- (I1) Given two morphisms $i \rightrightarrows j$ in I , there exists a morphism $j \rightarrow k$ so that its composite with these two morphisms are the same.
- (I2) Given two objects i and j in I , there exists an object k in I admitting morphisms $i \rightarrow k$ and $j \rightarrow k$.

Suppose objects in I form a set and I satisfies (I1) and (I2). For any covariant functor

$$F : I \rightarrow \mathcal{C}, \varinjlim_i F(i)$$

exists.

Remark.

Let (S, γ) be a pointed connected noetherian scheme. Let I be the opposite category of the category of pointed connected Galois étale covering spaces of (S, γ) . For any $i \in \text{obl}$, denote by (X_i, α_i) the corresponding pointed connected Galois étale covering space of (S, γ) . By Theorem 1(2) and Theorem 5, I satisfies the conditions (I1) and (I2), so I admits direct limit.

Definition. (Étale fundamental group)

We define the fundamental group of (S, γ) to be

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Recall the second definition of the fundamental group in algebraic topology.

Note that $\text{Aut}(X_i/S)^\circ$ acts on the left on the set $\text{Hom}(X_i, X)$ for any object X in $\text{Et}(S)$. So $\pi_1(S, \gamma)$ acts on the left on the sets

$$\varinjlim_i \text{Hom}_S(X_i, X) \cong F(X).$$

Then the action of $\pi_1(S, \gamma)$ on $F(X)$ factors through the finite quotient $\text{Aut}(X_i/S)^\circ$. Put the discrete topology on $\text{Aut}(X_i/S)^\circ$, and put the product topology on $\pi_1(S, \gamma) = \varprojlim_i \text{Aut}(X_i/S)^\circ$. Then $\pi_1(S, \gamma)$ acts continuously on the discrete finite set $F(X)$.

Theorem.

Let K be a field, Ω a separably closed field containing K , $\gamma : \text{Spec } \Omega \rightarrow \text{Spec } K$ the corresponding geometric point, and K_s the separable closure of K contained in Ω . Then we have a canonical isomorphism

$$\pi_1(\text{Spec } K, \gamma) \cong \text{Gal}(K_s/K).$$

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Proof.

Let $\{K_i\}$ be the family of finite Galois extensions of K contained in Ω and let $\alpha_i : \text{Spec } \Omega \rightarrow \text{Spec } K_i$ be the corresponding geometric points. Then $\{(\text{Spec } K_i, \alpha_i)\}$ is cofinal in the category of pointed connected Galois étale covering spaces of $(\text{Spec } K, \gamma)$. We have $\text{Aut}(\text{Spec } K_i / \text{Spec } K)^\circ = \text{Gal}(K_i/K)$. So we have

$$\pi_1(\text{Spec } K, \gamma) \cong \varprojlim_i \text{Gal}(K_i/K) \cong \text{Gal}(K_s/K).$$

Example.

- (1) $\pi_1(\mathbb{A}^1(\mathbb{C}) - \{0\}) \cong \varprojlim_i \mathbb{Z}/i\mathbb{Z} \cong \prod_p \mathbb{Z}_p = \hat{\mathbb{Z}}.$
- (2) $\pi_1(\operatorname{Spec}(\mathbb{F}_p)) \cong \hat{\mathbb{Z}}.$

Theorem.

Let S be a normal connected noetherian scheme, K its function field, Ω a separably closed field containing K , and K_s the separable closure of K in Ω . Denote by γ both the geometric point $\text{Spec } \Omega \rightarrow \text{Spec } K$ and the geometric point defined by the composite $\text{Spec } \Omega \rightarrow \text{Spec } K \rightarrow S$. Denote K_{ur} is the subfield of K_s generated by all finite separable extensions of K contained in K_s that are unramified over S . In particular, we have

$$\pi_1(S, \gamma) \cong \text{Gal}(K_{ur}/K).$$

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